

MSR-2 mmWave Multisensor

Compact mmWave presence and multi-sensor monitoring for Home Assistant

The Apollo MSR-2 is a compact mmWave multisensor built for Home Assistant and the successor to the popular MSR-1. Powered over USB-C and running ESPHome firmware, it is controlled entirely on your local network with no cloud connection or subscription required. At its core is an LD2410B 24 GHz radar for reliable presence detection, paired with light, UV, and optional CO₂ sensing plus onboard RGB and buzzer feedback.

Compared to the MSR-1, the MSR-2 adds a smaller form factor and a rear expansion slot that enables accessories such as a back-facing female USB-C port for clean installations. All firmware and CAD files are open source.

Key Specifications

Product	MSR-2 mmWave Multisensor (LD2410B)
Manufacturer	Apollo Automation (Versailles, KY, USA)
Presence sensor	HLK-LD2410B 24 GHz mmWave radar
Microcontroller	ESP32-C3
Firmware	ESPHome (Web Server v3) — open source
Connectivity	Wi-Fi (2.4 GHz) + Bluetooth
Power input	USB-C, standard 5 V wall adapter
Form factor	Compact — reduced size vs. MSR-1
Onboard feedback	RGB neopixel + piezo buzzer
Expansion	Front and rear expansion slots; GPIO header option
Local control	Yes — no cloud or subscription required
In the box	Assembled sensor with case; Apollo sticker

Onboard & Optional Sensors

Sensor	Function	Availability
HLK-LD2410B <i>Hi-Link</i>	24 GHz mmWave radar — single-target presence and still-person detection with tunable sensitivity	Standard
LTR-390UV <i>Liteon</i>	Ambient light (LUX) and ultraviolet (UV) sensing	Standard
SCD40 <i>Sensirion</i>	True CO ₂ (carbon dioxide) sensing	Optional
ESP32-C3-Mini <i>Espressif</i>	Bluetooth device / beacon tracking (YAML config required)	Standard

The LD2410B excels at still-person detection — it can keep lights on while you sit motionless and supports extensive tuning to eliminate false positives. mmWave tuning guides and videos are available on the Apollo wiki.

Configurations & Pricing

Configuration	CO2 (SCD40)	Price (USD)
Base — mmWave, LUX/UV, RGB, buzzer, Bluetooth	—	\$37.99

Configuration	CO2 (SCD40)	Price (USD)
With CO2	Yes	\$57.99

Optional add-on mounts (sold separately): USB-C wall-plug mount and ball mount.

Features & Use Cases

- **mmWave presence detection:** instantly trigger lights on entry and keep them on while you sit still; place under a bed for occupancy-based alarm and morning automations.
- **Light & UV sensing:** use LUX to only switch on lights when a room is dark, and UV to gauge how much sun is coming through a window.
- **CO₂ awareness (optional):** detect elevated indoor CO₂ that can affect sleep, focus, and comfort; trigger ventilation.
- **Piezo buzzer:** audible alerts you won't miss — beep on high CO₂ or a door opening, or play simple tunes.
- **RGB notifications:** color-coded alerts — e.g. green for trash night, blue for mail.
- **Bluetooth tracking:** locate phones, tags, or beacons by proximity to the sensor (YAML configuration required).

Important Notes

- **Single-target tracking:** the LD2410B radar tracks one target at a time. It is well suited to room presence rather than counting multiple people.
- **Tuning recommended:** mmWave performance depends on placement and sensitivity settings; review the Apollo tuning guides for best results.
- **Firmware:** the device uses the ESPHome web interface; updating to Web Server v3 improves stability and appearance. Updates install wirelessly via OTA.

Documentation & Resources

- **Product page:** apolloautomation.com/products/msr-2
- **Wiki / setup guide:** wiki.apolloautomation.com/products/msr2
- **mmWave tuning videos:** wiki.apolloautomation.com/products/general/calibrating-and-updating/mmwave-videos
- **Source & CAD files:** github.com/ApolloAutomation/MSR-2
- **Community:** [Apollo Automation Discord](https://discord.com/invite/apolloautomation)
- **Sensor datasheets:** Hi-Link LD2410B, Liteon LTR-390UV, Sensirion SCD40, Espressif ESP32-C3-Mini.